

Dual-Channel Variational Closure, Covariant Completion, and a Reproducible SPARC Benchmark: A Res-Nova v1.5.0 Technical Assessment

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Abstract

We present the Res-Nova v1.5.0 technical assessment of a dual-channel variational action for modified gravity in the AQUAL/MOND family. **Core result (proved):** The single-channel arcsinh action is falsified on correspondence grounds [P]; the dual-channel potential $\mathcal{F}_{\text{dual}}(x) = \frac{1}{2}x^2 - x + \ln(1+x)$ yields exactly $\mu(x) = x/(1+x)$ with correct Newtonian ($\mu \rightarrow 1$) and deep-MOND ($\mu \rightarrow x$) limits, proved algebraically and machine-checked in Lean 4. **Relativistic completion (proved):** Pure RAQUAL is falsified by superluminal characteristic cones [P]; the disformal completion is falsified by GW170817 [P]; the dual-channel action embeds exactly into the Skordis–Złótnik (2021) relativistic MOND framework with $c_T = c_\gamma = c$ and $\gamma_{\text{PPN}} = 1$ preserved identically [P]. **Empirical benchmark (computed):** A properly bootstrapped a_0 measurement across 171 SPARC galaxies (3,375 kinematic points) yields $a_0 = (1.116 \pm 0.128_{\text{stat}} \pm 0.097_{\text{syst}}) \times 10^{-10} \text{ m s}^{-2}$ (14.4% total), consistent with $cH_0/(2\pi)$ at 0.46σ and MOND’s canonical value at 0.52σ ; these two anchors are not separated at $z = 0$. A matched-parameter comparison shows the dual-channel MOND tier requires 342 fewer free parameters than a cosmologically-constrained NFW halo model at comparable or better fit quality. **Formal verification:** 17 Lean 4 modules, 0 sorry, axioms restricted to {propext, Classical.choice, Quot.sound}. **Open items:** The a_0 -horizon identity, $\Omega_\Lambda = \ln 2$, and CMB/structure-growth consistency remain quarantined as [O] pending future derivation and data.

Keywords: Modified Newtonian Dynamics, Dual-Channel AQUAL, SPARC Galaxy Kinematics, Skordis–Złótnik, Lean 4 Formal Verification, Horizon Thermodynamics.

1 Introduction and Scope

The asymptotic flatness of galactic rotation curves and the empirical Baryonic Tully–Fisher Relation (BTFR) [1] have motivated extensive investigation into modified gravity, including MOND [2] and AQUAL [3]. Prior work in this research program (PAPER_01, now quarantined) proposed a single-channel arcsinh action whose Euler–Lagrange derivative yields $\text{arcsinh}(x)$, not the required rational $\mu(x)$, establishing a correspondence failure.

This v1.5.0 assessment supersedes all prior working papers. It systematically delineates: what is mechanically proved [P], what is directly computed [D], what is cited from literature [C], and what

remains open [O]. The governing standard is the Sovereign Epistemic Covenant; the authoritative version record is `EPISTEMIC_BOUNDARY_v1.5.0.md`; in any conflict between this manuscript and the ledger, the ledger governs.

2 Epistemic Classification and Reproducibility Policy

All claims are governed by four explicit tiers:

- **[P] Proved / Machine-Verified:** Mathematical propositions mechanically checked in Lean 4 / Mathlib. Lean establishes only that statements follow from their encoded definitions; physical theory selection is not certified.
- **[D] Direct Empirical / Computed:** Quantities evaluated from raw, SHA-256-authenticated observational data using auditable, version-controlled scripts.
- **[C] Cited External Literature:** Established results from peer-reviewed sources.
- **[O] Open / Quarantined:** Conjectured boundary conditions, unproved physical mechanisms, or claims awaiting future derivation or data.

The complete claim-by-claim audit matrix is maintained in `CLAIM_EVIDENCE_LEDGER.md` and `EPISTEMIC_BOUNDARY_v1.5.0.md`. A human-readable referee index is in `FOR_REFEREES.md`.

3 No-Go Theorem for the Single-Channel Action (PAPER_01 Quarantine)

The single-channel AQUAL potential $\mathcal{F}_{\text{single}}(x) = x \operatorname{arcsinh}(x) - \sqrt{1+x^2}$ yields $\mathcal{F}'(x) = \operatorname{arcsinh}(x)$. This function satisfies $\operatorname{arcsinh}(x) \rightarrow 0$ as $x \rightarrow 0$ and $\operatorname{arcsinh}(x) \rightarrow \infty$ as $x \rightarrow \infty$ — the *inverted* MOND correspondence limits. The single-channel branch is therefore correspondence-false and is permanently quarantined as PAPER_01 [P]. This result is machine-certified in `05_lean_formalization/DualChannel`.

4 Dual-Channel Variational Closure

4.1 The Dual-Channel Action

Balancing a bulk kinetic flux channel against a horizon relative-entropy dissipation channel [**motivation, not a theorem; see O3**] isolates the unique two-channel potential:

$$\mathcal{F}_{\text{dual}}(x) = \frac{1}{2}x^2 - x + \ln(1+x), \quad x \equiv \frac{|\nabla\Phi|}{a_0} \geq 0. \quad (1)$$

4.2 Variational Derivation

Differentiating:

$$\mu(x) \equiv \mathcal{F}'_{\text{dual}}(x) = x - 1 + \frac{1}{1+x} = \frac{x}{1+x}. \quad [\mathbf{P}] \quad (2)$$

The asymptotic limits are exact:

$$\lim_{x \rightarrow 0} \mu(x) = x + O(x^2) \quad (\text{deep-MOND / BTFR}), \quad [\mathbf{P}] \quad (3)$$

$$\lim_{x \rightarrow \infty} \mu(x) = 1 - \frac{1}{x} + O(x^{-2}) \quad (\text{Newtonian}), \quad [\mathbf{P}] \quad (4)$$

Both limits and the full algebraic identity are machine-certified in Lean 4 (`DualChannelDerivation.lean`, `PPNLimits.lean`).

4.3 Kinetic Convexity and Stability

The second derivative:

$$\mathcal{F}_{\text{dual}}''(x) = \frac{x(x+2)}{(1+x)^2} > 0 \quad \forall x > 0. \quad [\mathbf{P}] \quad (5)$$

Strict convexity proves absence of tachyonic and gradient instabilities in the quasistatic Poisson field (`RelativisticStability.lean`).

5 Effective Weak-Field Model

In the weak-field, non-relativistic limit, the modified gravitational potential satisfies the AQUAL Poisson equation [3]:

$$\nabla \cdot \left[\mu \left(\frac{|\nabla \Phi|}{a_0} \right) \nabla \Phi \right] = 4\pi G \rho_{\text{bar}}, \quad (6)$$

with $\mu(x) = x/(1+x)$ [P]. Under spherical symmetry this reduces to the algebraic relation $g \cdot \mu(g/a_0) = g_{\text{bar}}$, giving the closed-form acceleration:

$$g = g_{\text{bar}} \left[\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{a_0}{g_{\text{bar}}}} \right]. \quad (7)$$

6 Acceleration-Scale Measurement and Open Horizon Identity

6.1 Fitted Measurement

A bootstrap-over-galaxies a_0 measurement on 171 SPARC galaxies (3,375 kinematic points, `A0_MEASUREMENT.json`) yields:

$$a_0 = (1.116 \pm 0.128_{\text{stat}} \pm 0.097_{\text{syst}}) \times 10^{-10} \text{ m s}^{-2} \quad (14.4\% \text{ total}). \quad [\mathbf{D}] \quad (8)$$

The horizon anchor $cH_0/(2\pi) \approx 1.044 \times 10^{-10} \text{ m s}^{-2}$ (Planck $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$) lies 0.46σ away; MOND’s canonical 1.2×10^{-10} lies 0.52σ away. These two anchors are not separated at $z = 0$ within current systematics.

Superseded value (do not cite): An earlier in-sample MAP run reported $a_0 = (9.433 \pm 0.050) \times 10^{-11}$. That error bar treated 3,391 radial points as independent and leaked the global a_0 into every cross-validation test fold. It is permanently superseded by Eq. (8).

6.2 Horizon Identity — Open Problem O1

Thermal equilibrium between the de Sitter Gibbons–Hawking temperature [4] and the Unruh temperature [5] yields $a = cH_0$; the universal KMS 2π factors cancel. The additional $1/(2\pi)$ divisor in $a_0 = cH_0/(2\pi)$ is therefore an **open boundary normalization** [O] (O1), not a first-principles derivation. Closing O1 requires an $a_0(z)$ measurement at high redshift, which does not yet exist.

7 SPARC Benchmark: Data, Methods, and Matched-Parameter Comparison

7.1 Observational Sample

We use the complete Spitzer Photometry and Accurate Rotation Curves (SPARC) database [6]: 175 disk galaxies, 3,391 kinematic observations. Raw data SHA-256: `e76e6752164b...c05f96d095456e067bdd5c6da5`. The a_0 measurement sub-sample is 171 galaxies / 3,375 points (4 galaxies excluded for insufficient data coverage).

7.2 Specification Definitions

- **GOD Tier 0 (Zero-Parameter):** $\Upsilon_{\text{disk}} = \Upsilon_{\text{bulge}} = f_d = 1.0$ fixed; a_0 from horizon formula. Tests the a_0 source hypothesis directly.
- **MOND Tier 0:** Same, with $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$ from literature [1].
- **GOD/MOND Tier 1 ($N_{\text{par}} = 374$):** 174 Υ_{disk} + 199 Υ_{bulge} (where applicable) + 1 global a_0 under $\mathcal{N}(0.5, 0.125^2)$ / $\mathcal{N}(0.7, 0.175^2)$ priors.
- **NFW Free- c ($N_{\text{par}} = 716$):** Per-galaxy NFW halo concentration c free. 97/171 railed at $c = 1$; not a fair Λ CDM proxy.
- **NFW Cosmological- c Prior ($N_{\text{par}} = 716$):** Concentration c constrained to cosmological mass-concentration relation. This is the fair Λ CDM-comparable row.

7.3 Matched-Parameter Results

Specification	Free Params	Median χ^2_{data}/N_g	Aggregate	Role
GOD Tier 0 (horizon a_0 , fixed M/L) [D]	0	9.20	—	Tests a_0 source hypothesis
MOND Tier 0 (literature a_0) [D]	0	11.35	—	Zero-param control
GOD Tier 1 [D]	374	2.95	—	Dual-channel μ, horizon a_0
MOND Tier 1 [D]	374	2.89	—	Dual-channel μ , fitted a_0
NFW Free c [D]	716	1.92	—	97/171 railed at $c = 1$; not Λ CDM
NFW Cosmological-c Prior [D]	716	5.62	—	Fair ΛCDM row; 342 extra params vs GOD

Table 1: Matched-parameter SPARC benchmark on 171 galaxies. GOD Tier 1 achieves median $\chi^2/N_g = 2.95$ with 374 parameters; the fair cosmological NFW model requires 342 additional parameters (716 total) for median $\chi^2/N_g = 5.62$. Parameter counts and fit results frozen in `PARAMETER_LEDGER.json` and `NFW_CONSTRAINED.json` (`VERIFICATION_RUN_003`).

7.4 Pre-Registered Falsification Contract

Pre-Registered Interpretation Contract

1. **Closure Test (A_4):** If the canonical 374-parameter GOD Tier 1 aggregate χ^2/dof underperforms the legacy μ_{simple} zero-parameter baseline in a like-for-like test, the derived closure is falsified.
2. **Scale Test (F_3):** Any persistent tension in fitted a_0 vs. horizon anchor exceeding 3σ falsifies the horizon-normalization narrative.
3. **No Silent Spec Changes:** Sample, error models, and priors frozen across all runs; changes must be versioned in the ledger.
4. **Parameter Ledger:** All parameter counts reported against identical galaxy samples. NFW comparisons must use cosmological- c priors to constitute a fair test.

8 4D Covariant Completion: Obstructions and Skordis–Złośnik Embedding

8.1 Pure RAQUAL Superluminality (Falsified)

In static galaxy halos with spacelike gradient ($x > 0$), the perturbation speed parallel to the gradient for any RAQUAL/k-essence action is:

$$c_{\parallel}^2 = 1 + \frac{2y\mathcal{F}''(y)}{\mathcal{F}'(y)} \Big|_{y=x^2} = 1 + \frac{1}{1+x} > 1 \quad \forall x > 0. \quad [\mathbf{P}] \quad (9)$$

Pure RAQUAL is **falsified** on superluminality grounds. Machine-certified in `CovariantCompletion.lean`.

8.2 Disformal Completion (Falsified by GW170817)

The disformal metric $\tilde{g}_{\mu\nu} = e^{-2\phi}g_{\mu\nu} - 2\sinh(2\phi)u_\mu u_\nu$ yields photon speed $c_\gamma = e^{2\phi}$ on $\tilde{g}$ and graviton speed $c_T = 1$ on g . The speed ratio:

$$\frac{c_T}{c_\gamma} = e^{-2\phi} = 1 \iff \phi = 0. \quad (10)$$

For deep-MOND halos ($\phi \sim 10^{-6}$) or cosmological backgrounds ($\phi \sim 0.1$), $|e^{-2\phi} - 1| \gg 10^{-15}$, violating the GW170817/GRB 170817A multi-messenger bound by 9–14 orders of magnitude. The disformal completion $B(\phi) = -2\sinh(2\phi)$ is **falsified** $[\mathbf{P}]$. Machine-certified in `TensorSpeed.lean`.

8.3 Skordis–Złośnik Embedding (Viable, Proved)

The Skordis–Złośnik (2021) [7] action admits a free kinetic function $\mathcal{J}(\mathcal{Y})$ on the spacelike projected gradient $\mathcal{Y} \equiv g^{\mu\nu}\hat{\nabla}_\mu\phi\hat{\nabla}_\nu\phi/a_0^2$. Setting:

$$\mathcal{J}(\mathcal{Y}) = \frac{1}{2}\mathcal{Y} - \sqrt{\mathcal{Y}} + \ln(1 + \sqrt{\mathcal{Y}}) \equiv \mathcal{F}_{\text{dual}}(\sqrt{\mathcal{Y}}) \quad (11)$$

yields $2\mathcal{J}'(\mathcal{Y}) = \sqrt{\mathcal{Y}}/(1 + \sqrt{\mathcal{Y}}) = \mu(\sqrt{\mathcal{Y}})$ $[\mathbf{P}]$.

Consequences (all machine-certified in `SkordisZlosnikEmbedding.lean`, `TensorSpeed.lean`):

- **GW170817:** Matter and photons couple to $g_{\mu\nu}$ directly; $c_T = c_\gamma = c$, so $|c_T/c_\gamma - 1| = 0 \leq 10^{-15}$ **identically** [P].
- **Lensing / PPN:** Dynamical vector A^μ shear cancels scalar trace discrepancies, maintaining $\Phi = \Psi$ ($\gamma_{\text{PPN}} = 1$, proved in `CovariantCompletion.lean`) without disformal distortion [P].
- **Preferred-frame bound:** $\alpha_1 = -2K$; solar system constraint $|\alpha_1| \lesssim 10^{-4}$ places $|K| \lesssim 5 \times 10^{-5}$ [P]/[D].
- **Ghost freedom:** $\mathcal{J}''(\mathcal{Y}) = \frac{1}{4\sqrt{\mathcal{Y}}(1+\sqrt{\mathcal{Y}})^2} > 0$ [P].
- **Cosmological decoupling:** On FLRW, $\hat{\nabla}_\mu \phi = 0 \implies \rho_\phi = 0$; the MOND kinetic sector contributes zero dynamical dark energy [P].

9 Lean 4 Formal Verification: Scope and Limits

The Lean 4 suite (`05.lean.formalization/`) comprises **17 modules**, 0 sorry, axioms `{propext, Classical.choice, Quot.sound}`, Lean `v4.33.0-rc1`, Mathlib pinned at `5eec30bc`. Gate: `verify_all_proofs.sh`.

Important provenance note: O6 — walked once in a clean worktree at 07185a6 (`lake exe cache get + 17/17 PASS, VERIFICATION_RUN_007`). Not yet demonstrated on a cold machine with empty host cache, and not yet a CI release gate. Artifact: `VERIFICATION_RUN_007/01.lean/` (transcript, per-target exit codes, toolchain and Mathlib revisions, SHA-256 checksums).

10 Open Problems

The following items are quarantined [O] in `OPEN_PROBLEMS_AND_TESTS.md`:

1. **O1 — a_0 -Horizon Identity:** $a_0 = \xi \cdot cH_0$. Closing requires $a_0(z)$ at high redshift; data does not yet exist.
2. **O2 — $\Omega_\Lambda = \ln 2$:** An asymptotic entropy boundary condition, not a dynamical prediction. A full curved-spacetime covariant derivation has not been written.
3. **O3 — $\mu(x)$ Physical Mechanism:** The algebraic closure is proved; the physical origin of the dual-channel balance (bulk kinetic flux vs. horizon dissipation) is a motivated conjecture.
4. **O4 — CMB and Structure Growth on FLRW:** FLRW decoupling is proved in the quasistatic limit. The full nonlinear perturbation treatment (acoustic oscillations, matter power spectrum, CMB) is not done. This is the primary open science frontier.
5. **O5 — Cluster Lensing and Bullet Cluster:** MOND-family theories require additional physics for galaxy cluster mass budgets; applicability of this framework at cluster scales is untested.
6. **O6 — Clean-Worktree Lean Build:** Walked once in a clean worktree at 07185a6 (`lake exe cache get + 17/17 PASS, VERIFICATION_RUN_007`). Not yet demonstrated on a cold machine with empty host cache, and not yet a CI release gate.

Module	What is proved [P] / What is not [O]
AXIOMS.V2.lean	<i>Assumption declarations.</i> States the five physical axioms A1–A5 as Lean type-classes. Its two theorems are a field projection and the positivity of $\sqrt{g_{\text{bar}}a_0}$; neither derives an axiom [O].
CosmologicalSector.lean	$0 < \ln 2 < 1$ and matter-density bounds [P]. $\Omega_\Lambda + \Omega_m = 1$ holds by construction, Ω_m being defined as $1 - \Omega_\Lambda$; it is not a flatness result. $\Omega_\Lambda = \ln 2$ is a definition, not a derivation [O].
CovariantCompletion.lean	RAQUAL superluminality; $\gamma_{\text{PPN}} = 1$ (<code>disformal_gamma_ppn_unity</code>); preferred-frame parameters zero; FLRW decoupling $\Rightarrow \rho_\phi = 0$ [P], conditional on the stated embedding.
DeSitterExtremal.lean	Horizon lapse vanishing at $r = 1/H$; flat limit; $cH/(2\pi) > 0$ [P]. Does not derive $a_0 = cH_0/(2\pi)$ [O].
DualChannelDerivation.lean	$\mathcal{F}'_{\text{dual}}(x) = x/(1+x)$; bounds $\mu(x) < 1$ and the deep-MOND bound [P]. Physical mechanism of channel balance [O].
GODActionKinematics.lean	Dual-channel polynomial identity; AQUAL simple- μ ratio; BTFR algebraic scaling [P].
ITActionClosure.lean	Polynomial τ -law / simple- μ relation; deep-MOND BTFR; flat rotation curve for $n = 2$ [P]. Relativistic stability [O].
MuProjection.lean	Properties of $\mu(x)$; root uniqueness; power-law solves the dilaton EOM; exponential profile falsified [P]. Physical derivation of closure [O].
PPNLimits.lean	Solar-system PPN precision bounds at $x \geq 10^4$; Cassini radar delay satisfied at $x \geq 6 \times 10^7$ [P]. No asymptotic limit is formalized [O].
PrintAxioms.lean	<i>Diagnostic helper, not a physics module.</i> Emits <code>#print axioms</code> for the <code>CovariantCompletion</code> results. No proof content.
PrintAxiomsD8.lean	<i>Diagnostic helper, not a physics module.</i> Re-declares a subset of <code>TensorSpeed</code> in the same namespace and prints its axiom footprint; not an independent result.
RelativisticStability.lean	Kinetic convexity $\mathcal{F}'' > 0$ [P].
SOCasimirGenuine.lean	$\mathfrak{so}(n)$ Casimir eigenvalue from explicit generators [P]. Physical gauge symmetry [O].
SkordisZlosnikEmbedding.lean	Exact $\mathcal{J}(\mathcal{V})$ embedding; $c_T = c$; weak-field lensing $\Phi = \Psi$ [P].
SovereignRegularity.lean	Pointwise vorticity bounded by an assumed control field; <code>sovereign_regularity_theorem</code> restates that hypothesis. No integral and no PDE are formalized, so this is not the Beale–Kato–Majda criterion [8] [O].
TensorSpeed.lean	$c_{13} = 0$ identity; disformal falsification; $\alpha_1 = -2K$ [P].
YettParadigm.lean	Positivity of an <i>assumed</i> spectral gap $\kappa^2 \leq \lambda_1 - \lambda_0$; bounded ADCCL trajectories [P]. <code>chiral_phase_stable</code> projects its own hypothesis: $\theta \geq 0.70$ is assumed, not proved [O].

Table 2: Source inventory of all 17 Lean 4 modules, enumerated from the tracked contents of `05_lean_formalization/`. Three entries are diagnostic or assumption-declaring files rather than physics results. Physical-theory selection is never certified; Lean certifies only that mathematical statements follow from their encoded definitions, and not that assumptions carried as typeclass or structure fields are justified (see `THEORY_ASSUMPTION_AUDIT.md`).

11 Cosmological and Optical Extensions (Open Research Program)

The observer and cosmological sectors incorporate proposed phenomenological relations, all quarantined [O]:

- **Disformal Optical Metric:** $g_{\mu\nu}^{\text{opt}} = g_{\mu\nu} + \beta(\chi)\nabla_\mu\chi\nabla_\nu\chi$.
- **Spectral Redshift Jacobian:** $1 + z_{\text{obs}} = (1 + z_{\text{cosm}})\mathcal{J}(\chi)$.
- **Luminosity Distance Modulation:** $d_L^{\text{eff}}(z) = d_L(z)\sqrt{1 - \chi/\chi_{\text{crit}}}$.

- **Horizon Density:** $\Omega_\Lambda(z=0) = \ln 2$.

Full derivations from Maxwell’s equations on curved spacetime and confrontation with CMB/BAO remain open (O4).

12 Limitations and Falsifiability

The research program is subject to clear falsification criteria:

1. **CMB Power Spectrum:** If the Skordis–Złóśnik completion with $\mathcal{J} = \mathcal{F}_{\text{dual}}(\sqrt{\mathcal{Y}})$ cannot reproduce the Planck CMB acoustic peak positions and amplitudes, the covariant embedding is falsified as a complete cosmological theory.
2. **Structure Growth:** If the predicted matter power spectrum $P(k)$ is inconsistent with BOSS/eBOSS galaxy clustering, the model fails at cosmological scales.
3. **Cluster Mass Budgets:** Systematic failure on galaxy cluster rotation/lensing would restrict the framework to galactic scales only.
4. $a_0(z)$ **Tension:** A $> 3\sigma$ departure of fitted a_0 from $cH_0/(2\pi)$ at any redshift falsifies O1.
5. **Lean Build Failure:** Failure of `verify_all_proofs.sh` on a clean clone would retract all [P] claims pending repair.

13 Conclusion

The Res-Nova v1.5.0 assessment establishes the following proved and computed results: the dual-channel action $\mathcal{F}_{\text{dual}}(x)$ yields $\mu(x) = x/(1+x)$ with exact correspondence limits [P]; the single-channel arcsinh action is falsified [P]; pure RAQUAL and the disformal completion are falsified [P]; the dual-channel action embeds exactly into Skordis–Złóśnik with $c_T = c$, $\gamma_{\text{PPN}} = 1$, ghost freedom, and FLRW decoupling [P]; and the a_0 measurement from 171 SPARC galaxies is $1.116 \times 10^{-10} \pm 14.4\%$, consistent with the horizon anchor at 0.46σ [D].

The remaining open problems (O1–O6) are documented, bounded, and do not retract the proved core. The primary scientific frontier is O4: CMB and structure-growth consistency on FLRW backgrounds. This is the next derivation target in the Res-Nova research program.

Data and Code Availability

All raw data, Python scripts, Lean 4 modules, and verification manifests are archived in the repository at <https://github.com/Mega-Therion/Res-Nova>, verified artifact references VERIFICATION_RUN_003, VERIFICATION_RUN_005, VERIFICATION_RUN_006, VERIFICATION_RUN_007.

Declarations

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Conflict of Interest: None.

Author Contributions: R.W.Y. — theoretical derivation, Lean formalization, numerical computation, manuscript preparation. AI tools (Perplexity, Claude) contributed to draft generation and code review under human direction and audit.

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Appendix A: Lean 4 Source Inventory and Verification Provenance

The tracked Lean 4 sources under `05_lean_formalization/` comprise **17 modules**, enumerated in Table 2. This appendix records the headline declarations per module and the provenance of the verification run; it adds no claim beyond that table.

Verification provenance. O6 — walked once in a clean worktree at 07185a6 (`lake exe cache get` + 17/17 PASS, `VERIFICATION_RUN_007`). Not yet demonstrated on a cold machine with empty host cache, and not yet a CI release gate. The gate `verify_all_proofs.sh` elaborated all 17 declared targets with exit status 0, no `sorry` in proof code, and axiom footprints confined to `{propext, Classical.choice, Quot.sound}`, under Lean v4.33.0-rc1 and Mathlib 5eec30bc. The transcript, per-target exit codes, toolchain and Mathlib revisions, and SHA-256 checksums are archived in `VERIFICATION_RUN_007/01_lean/`.

What elaboration does and does not certify. A green gate run certifies that each statement follows from its encoded definitions. It does not certify that those definitions model the physical system, and it does not certify assumptions carried as typeclass fields, structure fields, or theorem hypotheses. The suite declares no Lean global `axiom`; that is a statement about `axiom` declarations only. See `THEORY_ASSUMPTION_AUDIT.md`.

Scope note on `SovereignRegularity.lean`. The module bounds pointwise vorticity by an assumed control field. It does not formalize an integral and does not formalize a partial differential equation, so it does not establish the Beale–Kato–Majda criterion [8], which concerns $\int_0^T \|\omega\|_\infty dt$. `sovereign_regularity_theorem` restates the module’s own hypothesis.

Appendix B: SPARC Metric Definitions

For each galaxy g with N_g observed kinematic radii R_i , observed velocities $V_{\text{obs},i}$, and observational uncertainties $\sigma_{V,i}$, the model velocity is:

$$V_{\text{model}}(R_i) = V_{\text{bar}}(R_i) \cdot \left[\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{a_0}{g_{\text{bar}}(R_i)}} \right]^{1/2}, \quad (12)$$

where $g_{\text{bar}}(R_i) = V_{\text{bar}}(R_i)^2/R_i$, with $V_{\text{bar}}^2(R) = |V_{\text{gas}}|V_{\text{gas}} + \Upsilon_{\text{disk}}|V_{\text{disk}}|V_{\text{disk}} + \Upsilon_{\text{bulge}}|V_{\text{bulge}}|V_{\text{bulge}}$.

Per-galaxy metric

$$\frac{\chi_{\text{data}}^2}{N_g} = \frac{1}{N_g} \sum_{i=1}^{N_g} \left(\frac{V_{\text{obs},i} - f_d V_{\text{model}}(f_d R_i)}{\sigma_{V,i}} \right)^2. \quad (13)$$

These are definitions only. The sample size, parameter count, and all fitted values are carried by the frozen artifacts named in Appendix C; they are deliberately not restated here, so that this appendix cannot drift from them.

Module	Headline declarations
<i>Proof modules</i>	
CosmologicalSector.lean	log_two_pos, log_two_lt_one, matter_density_bounds, spatial_flatness_sum
CovariantCompletion.lean	raqual_superluminal_obstruction, disformal_gamma_ppn_unity, preferred_frame_parameters_zero, flrw_projected_gradient_zero, no_dynamical_dark_energy_density
DeSitterExtremal.lean	desitter_lapse_horizon, expr_cH_over_2pi_pos, desitter_flat_limit, volume_law_weight_nonneg
DualChannelDerivation.lean	dual_channel_flux_algebra, mu_derived_inversion, mu_derived_deep_mond_upper_bound, mu_derived_newtonian_bound
GODActionKinematics.lean	dual_channel_poly_identity, aqual_simple_mu_ratio, btfr_algebraic_scaling
ITActionClosure.lean	tauLaw_eq_simple_mu_poly, btfr_deep_mond, enclosed_mass_n2_linear, flat_rotation_curve_n2
MuProjection.lean	mu_simple_eq_cos, quadratic_law_root_unique, powerLaw_solves_dilaton_eom, powerLaw_iterated_deriv, exp_profile_fails_cubic
PPNLimits.lean	fractional_deviation_eq, solar_system_precision_bound, cassini_radar_delay_satisfied
RelativisticStability.lean	first_derivative_pos, second_derivative_pos, ghost_free_convexity
SOCasimirGenuine.lean	gen_skew, sum_dg, casimir_defining_rep, gen_sq, casimir_scalar_eq
SkordisZlosnikEmbedding.lean	sz_aqual_reduction, dJ_dY_pos, sz_tensor_speed_luminal, sz_weak_field_lensing
SovereignRegularity.lean	chiral_iff_lipschitz_constant, lipschitz_implies_angle_modulus, bkm_no_blowup, sovereign_regularity_theorem
TensorSpeed.lean	c_T_sq_at_zero, maxwellian_c13_vanishes, foster_jacobson_alpha_1_eval, speed_ratio_unity_iff, gw170817_deviation_of_pos
YettParadigm.lean	ramanujan_yett_spectral_gap_pos, adccl_trajectory_bounded, chiral_phase_stable
<i>Assumption declarations — not physics results</i>	
AXIOMS_V2.lean	Typeclasses Axiom_A1–A5; derived_simple_mu_bounds, deep_mond_baryonic_scaling
<i>Diagnostic helpers — no proof content</i>	
PrintAxioms.lean	#print axioms for the CovariantCompletion results
PrintAxiomsD8.lean	#print axioms for the tensor-speed results

Table 3: Headline declarations in the 17 tracked Lean 4 source modules, enumerated from `git ls-files`. Epistemic status per module is given in Table 2; this table lists declaration names only.

Appendix C: Parameter Accounting and Cross-Validation

Superseded accounting — do not quote. Earlier revisions of this appendix reported an aggregate residual over $N_{\text{data}} = 3,391$ points with $N_{\text{par}} = 382$ and $\text{DOF}_{\text{nom}} = 3,009$, and a 5-fold cross-validation evaluated against *fixed* population priors. Both are superseded: items **D4.1** and **D4.2** of `EPISTEMIC_BOUNDARY_v1.5.0.md` record the method defects — radial points treated as independent, and a cross-validation that held a_0 fixed across folds and so was not a cross-validation.

Those numbers are retained here only to be labelled, and must not be quoted as current results.

Current accounting. The parameter ledger, sample definition, systematic budget, and honest per-fold cross-validation are carried by the frozen artifacts on `main`:

- `02_galaxy_dynamics/A0_MEASUREMENT.json` — working a_0 with the statistical and systematic budget (item D4.3).
- `02_galaxy_dynamics/PARAMETER_LEDGER.json` — tiered parameter counts and per-tier medians (items D4.7, D4.8).
- `02_galaxy_dynamics/NFW_CONSTRAINED.json` — concentration-prior comparison (item D4.9).
- `02_galaxy_dynamics/A0_ESTIMATE.json` — cross-validation retraining a_0 per fold (item D4.6).
- `02_galaxy_dynamics/SPARC_PARAMETER_BUDGET.md` — the budget derivation.

Where this appendix and `EPISTEMIC_BOUNDARY_v1.5.0.md` appear to differ, the boundary document governs.

Appendix D: Reproduction Commands

All paths are relative to the repository root. SPARC rotation-curve tables are not vendored; see `02_galaxy_dynamics/SPARC_DATA.md` for the official source and the checksum manifest.

```
# 1. Lean 4 formal suite (all 17 declared targets)
cd 05_lean_formalization
lake exe cache get          # fresh-clone path (see 06)
./verify_all_proofs.sh      # exits 0 only if every target elaborates cleanly

# 2. SPARC measurement and parameter ledger
#   Requires the rotation-curve tables; see 02_galaxy_dynamics/SPARC_DATA.md
cd 02_galaxy_dynamics
python3 a0_measure.py
python3 parameter_ledger.py

# 3. Manuscript PDF
pdflatex -interaction=nonstopmode final_manuscript.tex
bibtex   final_manuscript
pdflatex -interaction=nonstopmode final_manuscript.tex
pdflatex -interaction=nonstopmode final_manuscript.tex
```

Appendix E: Artifact Manifests

Earlier revisions printed a table of truncated SHA-256 prefixes, which cannot be verified. The full manifests are tracked in the repository and should be checked directly:

- `VERIFICATION_RUN_001/02_sparc_strict_135/RAW_DATA_MANIFEST.sha256` — SHA-256 of each of the 175 `*_rotmod.dat` inputs. Verify with `sha256sum -c` from the data directory.

- `VERIFICATION_RUN_003/01_lean/SHA256SUMS.txt` — checksums of the Lean gate transcript and its machine-readable result record.
- `05_lean_formalization/lake-manifest.json` — pinned Mathlib revision and transitive dependency revisions.
- `05_lean_formalization/lean-toolchain` — pinned Lean toolchain.